CBSE CLASS 12 MATHEMATICS

Complete Formula Handbook — 2025-26 Board Exam Edition

Updated as per OFFICIAL CBSE Syllabus (cbseacademic.nic.in)

Sample formulas: $A^{-1} = \frac{adj(A)}{|A|}$ $\int_a^b f(x)dx = F(b) - F(a)$

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Official CBSE Syllabus 2025-26 — Deleted Topics

Unit	Topics Covered	Marks
I: Relations & Functions	Relations & Functions + Inverse Trigonometric Functions	08
II: Algebra	Matrices + Determinants	10
III: Calculus	Continuity & Diff. + App.Deriv. + Integrals + App.Integrals + Diff.Equations	35
IV: Vectors & 3D	Vectors + Three Dimensional Geometry (Lines only)	14
V: Linear Programming	LPP — Graphical method, up to 3 non-trivial constraints	05
VI: Probability	Conditional Probability + Bayes' Theorem only	08
	TOTAL THEORY	80

🗑 REMOVED FROM 2025-26 SYLLABUS (Official CBSE)

- Ch 1: Composition of functions (gof), $(gof)^{-1} = f^{-1}og^{-1}$, Binary Operations
- Ch 4: Cramer's Rule — use Inverse Matrix method ($X = A^{-1}B$) only
- Ch 5: Rolle's Theorem — NOT in 2025-26 syllabus
- Ch 5: Lagrange's Mean Value Theorem (LMVT) — NOT in 2025-26 syllabus
- Ch 6: Tangents and Normals — REMOVED
- Ch 6: Approximations ($\Delta y \approx dy$) — NOT in syllabus
- Ch 9: Bernoulli's Equation — NOT in 2025-26 syllabus
- Ch 10: Scalar Triple Product — NOT in 2025-26 syllabus
- Ch 11: ENTIRE Planes section (angles, distances, foot of perp) — REMOVED; Only Lines remain
- Ch 13: Random Variables, Probability Distribution, Binomial Distribution, Mean & Variance — ALL REMOVED

CHAPTER 1

Relations & Functions

Types of relations | One-one | Onto | Bijective | Inverse functions

1.1 Concept Overview

Concept	Definition
Reflexive	$(a,a) \in R$ for all $a \in A$
Symmetric	$(a,b) \in R \Longrightarrow (b,a) \in R$
Transitive	$(a,b) \in R$ and $(b,c) \in R \Longrightarrow (a,c) \in R$
Equivalence	Reflexive + Symmetric + Transitive simultaneously
One-one (Injective)	$f(a) = f(b) \Longrightarrow a = b$ [distinct inputs → distinct outputs]
Onto (Surjective)	Range = Codomain (every $y \in B$ has a pre-image in A)
Bijective	One-one AND onto; ONLY bijective functions have an inverse

1.2 Core Formulas

► Number of relations on set with n elements = 2^{n^2}

► f^{-1} exists if and only if f is **bijective (one-one + onto)**

🗑 **REMOVED FROM 2025-26 SYLLABUS (Official CBSE)**

- Composition (gof), $(gof)^{-1} = f^{-1}og^{-1}$ — removed from Class 12 summative exam

1.3 Important Results

📌 **Equivalence Classes**

- Equivalence class $[a] = \{b \in A : (a,b) \in R\}$
- Equivalence classes partition the set (disjoint, union = whole set)
- Number of equivalence relations on $\{1,2,3\} = 5$ (Bell number B_3)
- Symmetric and transitive does NOT guarantee reflexive — always verify all three

⚠ **Common Mistakes**

- $f^{-1}(x)$ is NOT $1/f(x)$. Inverse function ≠ reciprocal.
- Onto: Range = Codomain exactly. If range ⊂ Codomain, f is INTO, not onto.
- Equivalence requires ALL THREE conditions — missing one = not equivalence.

1.4 Worked Example

Worked Example

Q: Show that f: $\mathbb{R} \to \mathbb{R}$, f(x) = 3x + 2 is bijective.

Step 1: One-one: Let $f(a) = f(b) \implies 3a+2 = 3b+2 \implies a = b$ ✓

Step 2: Onto: For any $y \in \mathbb{R}$, let $x = (y-2)/3 \in \mathbb{R}$, then $f(x) = y$ ✓

∴ **Answer: f is bijective.**

1.5 PYQ Triggers

Year	Board Exam Question
2024	Show f: $N \to N$, $f(n) = n+(-1)^n$ is bijective. [3 marks]
2023	Let R on N: $R=\{(a,b): a=b^2\}$. Check reflexive, symmetric, transitive. [3 marks]
2022	Check if f: $\mathbb{R}\to\mathbb{R}$, $f(x)=x^2$ is one-one and onto. [2 marks]
2021	Write all equivalence relations on A={1,2,3} containing (1,2) and (2,1). [3 marks]
2019	Show $f(x)=(4x+3)/(3x+4)$ is bijective on $\mathbb{R}-\{-4/3\}\to\mathbb{R}-\{4/3\}$. Find f^{-1}. [5 marks]

CHAPTER 2

Inverse Trigonometric Functions

Domain & Range | Principal Values | All identities | Simplification

2.1 Domain, Range & Principal Value Branch

Function	Domain	Principal Value Range	Graph
$\sin^{-1}x$	[−1, 1]	[−π/2, π/2]	Increasing, through origin
$\cos^{-1}x$	[−1, 1]	[0, π]	Decreasing
$\tan^{-1}x$	ℝ	(−π/2, π/2)	Increasing, asymptotes
$\cot^{-1}x$	ℝ	(0, π)	Decreasing, asymptotes
$\sec^{-1}x$	ℝ−(−1,1)	[0,π]−{π/2}	Gap at π/2
$\text{cosec}^{-1}x$	ℝ−(−1,1)	[−π/2,π/2]−{0}	Gap at 0

2.2 Complementary Identities

- $sin^{-1}x + cos^{-1}x = \frac{\pi}{2}$ $for\, x \in [-1, 1]$
- $tan^{-1}x + cot^{-1}x = \frac{\pi}{2}$ $for\, x \in \mathbb{R}$
- $sec^{-1}x + cosec^{-1}x = \frac{\pi}{2}$ $for\, |x| \geq 1$

2.3 Negative Argument Identities

- $\sin^{-1}(-x) = -\sin^{-1}x$
- $\cos^{-1}(-x) = \pi - \cos^{-1}x$ ← *NOT $-\cos^{-1}x$!*
- $\tan^{-1}(-x) = -\tan^{-1}x$
- $\cot^{-1}(-x) = \pi - \cot^{-1}x$
- $\sec^{-1}(-x) = \pi - \sec^{-1}x$
- $\text{cosec}^{-1}(-x) = -\text{cosec}^{-1}x$

2.4 Addition/Subtraction Formulas ★ Most Tested

- $tan^{-1}x + tan^{-1}y = tan^{-1}\left(\frac{x+y}{1-xy}\right)$ $if\, xy < 1$
- $tan^{-1}x + tan^{-1}y = \pi + tan^{-1}\left(\frac{x+y}{1-xy}\right)$ $if\, x > 0, y > 0, xy > 1$
- $2tan^{-1}x = sin^{-1}\frac{2x}{1+x^2} = cos^{-1}\frac{1-x^2}{1+x^2} = tan^{-1}\frac{2x}{1-x^2}$

2.5 Reciprocal & Conversion Identities

- $\sin^{-1}(1/x) = \text{cosec}^{-1}x$ for $|x| \geq 1$
- $\cos^{-1}(1/x) = \sec^{-1}x$ for $|x| \geq 1$
- $\tan^{-1}(1/x) = \cot^{-1}x$ for $x > 0$
- $\tan^{-1}(1/x) = -\pi + \cot^{-1}x$ for $x < 0$
- $\sin^{-1}x = \cos^{-1}\sqrt{(1-x^2)} = \tan^{-1}(x/\sqrt{(1-x^2)})$ for $x \in [0,1]$

Finding Principal Value — Step by Step

- $\sin^{-1}(\sqrt{3}/2)$: Find $\theta \in [-\pi/2, \pi/2]$ with $\sin\theta = \sqrt{3}/2 \rightarrow \theta = \pi/3$
- $\cos^{-1}(-1/2)$: Find $\theta \in [0, \pi]$ with $\cos\theta = -1/2 \rightarrow \theta = 2\pi/3$ (NOT $-\pi/3$!)
- $\tan^{-1}(-1)$: Find $\theta \in (-\pi/2, \pi/2)$ with $\tan\theta = -1 \rightarrow \theta = -\pi/4$
- $\sec^{-1}(-2)$: $\cos\theta = -1/2$, $\theta \in [0,\pi] \rightarrow \theta = 2\pi/3$

⚠ Common Mistakes

- $\cos^{-1}(-x) = \pi - \cos^{-1}x$, NOT $-\cos^{-1}x$. Cosine inverse is NOT odd.
- Check $xy < 1$ or $xy > 1$ BEFORE applying $\tan^{-1}$ addition formula.
- $\sin^{-1}(\sin 5\pi/6) \neq 5\pi/6$. Since $5\pi/6 \notin [-\pi/2,\pi/2]$: $\sin(5\pi/6)=1/2 \rightarrow$ answer$=\pi/6$
- $\sin^{-1}x \neq 1/\sin x$ — most common conceptual error.

2.6 Worked Example

Worked Example

Q: Evaluate: $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$

Step 1: $\tan^{-1}(\sqrt{3})$: $\theta \in (-\pi/2,\pi/2)$ with $\tan\theta = \sqrt{3} \rightarrow \theta = \pi/3$

Step 2: $\sec^{-1}(-2)$: $\cos\theta = -1/2$, $\theta \in [0,\pi] \rightarrow \theta = 2\pi/3$

Step 3: Result $= \pi/3 - 2\pi/3 = -\pi/3$

∴ **Answer: $-\pi/3$**

2.7 PYQ Triggers

Year	Board Exam Question
2024	Write value of $\cos^{-1}(\cos(7\pi/6))$. [1 mark]
2023	Evaluate: $\sin(\pi/3 - \sin^{-1}(-1/2))$. [2 marks]
2022	Prove: $\tan^{-1}(1/2) + \tan^{-1}(2/11) = \tan^{-1}(3/4)$. [3 marks]
2021	Prove: $3\sin^{-1}x = \sin^{-1}(3x-4x^3)$ for $x\in[-1/2,1/2]$. [3 marks]
2019	Find: $\tan^{-1}[2\sin(2\cos^{-1}(\sqrt{3}/2))]$. [2 marks]

CHAPTER 3

Matrices

Types | Operations | Transpose | Symmetric & Skew | Invertible Matrices

3.1 Types of Matrices

Type	Condition
Row Matrix	Order 1×n
Column Matrix	Order m×1
Square Matrix	m = n
Zero Matrix O	All elements = 0
Identity Matrix I	a_{ij} = 1 if i=j, else 0
Diagonal	a_{ij} = 0 for i ≠ j
Symmetric	A = A^T (i.e. $a_{ij} = a_{ji}$)
Skew-Symmetric	A = $-A^T$; all diagonal entries = 0

3.2 Operations

- Multiplication: $(AB)_{ij} = \Sigma a_{ik} \cdot b_{kj}$ [A is m×n, B is n×p → AB is m×p]
- AB is defined ONLY when cols(A) = rows(B)
- AB ≠ BA (NOT commutative in general)
- A(BC) = (AB)C | A(B+C) = AB+AC | AI = IA = A
- AB = O does NOT imply A=O or B=O
- AB = AC does NOT imply B=C (no cancellation law)

3.3 Transpose — Properties

- $(AB)^T = B^T A^T \leftarrow REVERSAL\ LAW$ *(Most common exam trap!)*
- $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$ [Symmetric part + Skew-symmetric part]
- $(A^T)^T = A$ | $(A+B)^T = A^T+B^T$ | $(kA)^T = kA^T$
- $(A+A^T)$ is always symmetric | $(A-A^T)$ is always skew-symmetric
- For skew-symmetric matrix of odd order: |A| = 0

3.4 Invertible Matrices

- A is invertible iff |A| ≠ 0 (non-singular)
- $A \cdot A^{-1} = A^{-1} \cdot A = I$ (inverse is unique)
- $(AB)^{-1} = B^{-1} \cdot A^{-1}$ [Reversal Law for inverse]
- $(A^T)^{-1} = (A^{-1})^T$

Finding A^{-1} by Row Reduction

- Step 1: Write augmented matrix [A | I].
- Step 2: Apply elementary row operations ($R_i \leftrightarrow R_j$, $R_i \to kR_i$, $R_i \to R_i + kR_j$).
- Step 3: Reduce LEFT side to identity I. RIGHT side becomes A^{-1}.
- If left side cannot become I: A is singular, A^{-1} does not exist.

⚠ Common Mistakes

- $(AB)^T \neq A^T B^T$. ALWAYS reverse the order: $(AB)^T = B^T A^T$
- AB = AC does NOT imply B = C. No cancellation law.
- 2×3 × 3×4 = 2×4 matrix, NOT 3×3.

3.5 Worked Example

Worked Example

Q: Express A = [[2,3],[−1,4]] as sum of symmetric and skew-symmetric matrix.

Step 1: A^T = [[2,−1],[3,4]]

Step 2: Symmetric P = ½($A+A^T$) = ½[[4,2],[2,8]] = [[2,1],[1,4]]

Step 3: Skew Q = ½($A-A^T$) = ½[[0,4],[−4,0]] = [[0,2],[−2,0]]

Step 4: Check: P + Q = [[2,3],[−1,4]] = A ✓

∴ **Answer: A = [[2,1],[1,4]] + [[0,2],[−2,0]]**

3.6 PYQ Triggers

Year	Board Exam Question
2024	If [[0,x+2],[x−5,0]] is skew-symmetric, find x. [1 mark]
2023	If A=[[2,−1],[3,4]], B=[[1,0],[−1,2]], verify $(AB)^T=B^T A^T$. [3 marks]
2022	Using elementary operations, find A^{-1} for A=[[1,3,−2],[−3,0,−1],[2,1,0]]. [5 marks]
2021	Show A=[[cosθ,sinθ],[−sinθ,cosθ]] satisfies $A^T A=I$. [2 marks]

CHAPTER 4

Determinants

Expansion | Properties | Cofactors | Adjoint | Inverse | Solving Equations

4.1 Determinant Formulas

2×2 Matrix:

- ► $|a \ b \ c \ d| = ad - bc$
- ► 3×3: Expand along row/column with MOST zeros to save time
- ► Cofactor: $C_{ij} = (-1)^{i+j} \times M_{ij}$ [M_{ij} = minor = det after deleting row i, col j]

4.2 Key Properties

Property	Statement
Rows ↔ Columns	$|A| = |A^T|$
Interchange	Swap two rows/cols → sign of |A| changes
Identical rows	|A| = 0 if any two rows (or cols) are identical
Zero row	|A| = 0 if any row/col is all zeros
Scalar k × row	Multiplying one row by k multiplies |A| by k
|kA|	$|kA| = k^n|A|$ (n = order of matrix)
Row operation	Adding k × another row leaves |A| unchanged
Product rule	|AB| = |A|·|B|
Triangular	|A| = product of all diagonal elements

4.3 Adjoint & Inverse

- ► $A^{-1} = \frac{adj(A)}{|A|} - valid\ ONLY\ when\ |A| \neq 0$
- ► $|adj\ A| = |A|^{n-1}\ for\ matrix\ of\ order\ n$
- ► adj(A): replace each element by its cofactor, then TRANSPOSE
- ► For 2×2: adj[[a,b],[c,d]] = [[d,−b],[−c,a]] [swap diagonal, negate off-diag]
- ► A·adj(A) = adj(A)·A = |A|·I

🗑 **REMOVED FROM 2025-26 SYLLABUS (Official CBSE)**

- Cramer's Rule — NOT in 2025-26. Use Inverse Matrix method ($X = A^{-1}B$) only.

4.4 Solving Linear Equations (AX = B method)

- ► Write AX = B (A = coefficient matrix, X = variable column, B = constant column)
- ► If |A| ≠ 0: Unique solution $X = A^{-1}B$
- ► If |A| = 0 and (adj A)B ≠ O: NO solution (inconsistent)
- ► If |A| = 0 and (adj A)B = O: Infinite solutions (consistent, dependent)

4.5 Area of Triangle

► Area = $\frac{1}{2}|det[x_1\, y_1\, 1\ /\ x_2\, y_2\, 1\ /\ x_3\, y_3\, 1]|$ Points collinear iff det = 0

⚠ Common Mistakes

- Cofactor $C_{ij} = (-1)^{(i+j)} \times M_{ij}$. Forgetting the sign ruins the answer.
- adj(A) requires TRANSPOSING the cofactor matrix — very often skipped.
- $(AB)^{-1} = B^{-1}A^{-1}$, NOT $A^{-1}B^{-1}$. Always reverse the order.

4.6 Worked Example

Worked Example

Q: Solve using inverse matrix: x+y+z=6, 2x+y+3z=14, x+4y+9z=36.

Step 1: A=[[1,1,1],[2,1,3],[1,4,9]], $B=[6,14,36]^T$
Step 2: |A| = 1(9−12)−1(18−3)+1(8−1) = −3−15+7 = −11 ≠ 0
Step 3: Find A^{-1} via cofactors. $X = A^{-1}B$

∴ **Answer: x = 1, y = 2, z = 3**

4.7 PYQ Triggers

Year	Board Exam Question
2024	Using cofactors of 2nd row, evaluate |5,3,8/2,0,1/1,2,3|. [3 marks]
2023	If A=[[1,−1,0],[2,3,4],[0,1,2]], find adjA and A^{-1}. [5 marks]
2022	Find k for which 2x+ky=1, 3x−5y=7 has unique solution. [2 marks]
2021	If A=[[2,−3],[3,4]], show $A^2-6A+17I=0$. Find A^{-1}. [5 marks]
2019	Find area of △ with vertices (3,8),(−4,2),(5,1). [2 marks]

CHAPTER 5

Continuity & Differentiability

Continuity | Differentiability | All derivative formulas | Chain rule | Second order

5.1 Continuity

► f is continuous at x=a iff: $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a)$

Continuity Test for Piecewise Functions

- Step 1: Find LHL = $\lim(x\to a^-)f(x)$ [put x = a−h, $h\to 0^+$]
- Step 2: Find RHL = $\lim(x\to a^+)f(x)$ [put x = a+h, $h\to 0^+$]
- Step 3: Find f(a) directly from the formula
- Step 4: LHL = RHL = f(a) → Continuous; otherwise → Discontinuous

REMOVED FROM 2025-26 SYLLABUS (Official CBSE)

- Rolle's Theorem — NOT in 2025-26 CBSE Class 12 syllabus
- Lagrange's Mean Value Theorem (LMVT) — NOT in 2025-26 syllabus

5.2 Differentiability

► LHD = $\lim_{h\to 0^+} \frac{f(a-h)-f(a)}{-h}$ RHD = $\lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h}$

► Differentiable ⟹ Continuous. But Continuous ⇏ Differentiable

► f(x) = |x| is continuous at x=0 but NOT differentiable at x=0

5.3 Derivative Formulas — Complete Table

f(x)	f '(x)	f(x)	f '(x)
x^n	$n\cdot x^{n-1}$	e^x	e^x
a^x	$a^x \cdot \ln a$	ln x	1/x
log_a x	1/(x·ln a)	√x	1/(2√x)
sin x	cos x	cos x	−sin x
tan x	$\sec^2 x$	cot x	$-\text{cosec}^2 x$
sec x	sec x · tan x	cosec x	−cosec x · cot x
$\sin^{-1}x$	$1/\sqrt{1-x^2}$	$\cos^{-1}x$	$-1/\sqrt{1-x^2}$
$\tan^{-1}x$	$1/(1+x^2)$	$\cot^{-1}x$	$-1/(1+x^2)$
$\sec^{-1}x$	$1/(\lvert x\rvert\sqrt{x^2-1})$	$\text{cosec}^{-1}x$	$-1/(\lvert x\rvert\sqrt{x^2-1})$

5.4 Rules of Differentiation

► **Chain Rule:** $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

► Product Rule: $\frac{d}{dx}[uv] = u \cdot v' + v \cdot u'$

- Quotient Rule: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v{\cdot}u\prime - u{\cdot}v\prime}{v^2}$
- Parametric: $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$
- Implicit: differentiate both sides w.r.t. x, collect dy/dx terms, solve
- Logarithmic: when y = [f(x)]^g(x), take ln both sides first

5.5 Second Order Derivatives

- $y_2 = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$
- Parametric d^2y/dx^2 = [d(dy/dx)/dt] / (dx/dt) — NOT $(d^2y/dt^2)/(d^2x/dt^2)$!

⚠ Common Mistakes

- d/dx[a^x] = a^x · ln a, NOT just a^x
- Parametric 2nd derivative: NEVER write $(d^2y/dt^2)/(d^2x/dt^2)$. Use [d(dy/dx)/dt]/(dx/dt).
- In log differentiation: take ln FIRST, then differentiate.
- d/dx|x| = x/|x| = sgn(x); undefined at x=0.

5.6 Worked Example

Worked Example

Q: If y = $\sin^{-1}(2x\sqrt{1-x^2})$, find dy/dx.

Step 1: Let x = sinθ. Then: $2x\sqrt{1-x^2}$ = 2sinθcosθ = sin2θ

Step 2: y = $\sin^{-1}$(sin2θ) = 2θ = $2\sin^{-1}x$

Step 3: dy/dx = 2 · $1/\sqrt{1-x^2}$

∴ **Answer: dy/dx = $2/\sqrt{1-x^2}$**

5.7 PYQ Triggers

Year	Board Exam Question
2024	If $y=(\sin x)^x+\sin^{-1}\sqrt{x}$, find dy/dx. [3 marks]
2023	Differentiate: y = x^x + (sinx)^(cosx). [3 marks]
2022	Find d^2y/dx^2 if x=a(θ+sinθ), y=a(1−cosθ). [3 marks]
2021	If $y=\log[x+\sqrt{x^2+a^2}]$, prove $(x^2+a^2)y_2+xy_1=0$. [3 marks]
2019	Find dy/dx if $(\cos x)^y=(\cos y)^x$. [3 marks]

CHAPTER 6

Application of Derivatives

Rate of Change | Increasing/Decreasing | Maxima & Minima

🗑 REMOVED FROM 2025-26 SYLLABUS (Official CBSE)

- Tangents and Normals — REMOVED from 2025-26 CBSE syllabus
- Approximations ($\Delta y \approx dy$) — NOT in 2025-26 syllabus

6.1 Rate of Change

- ► Rate of change of y w.r.t. x = $\frac{dy}{dx}$
- ► When both x,y depend on time t: $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$

6.2 Increasing & Decreasing Functions

- ► f is strictly increasing on (a,b) if $f'(x) > 0$ for all $x \in (a,b)$
- ► f is strictly decreasing on (a,b) if $f'(x) < 0$ for all $x \in (a,b)$
- ► Critical points: where $f'(x) = 0$ or $f'(x)$ undefined
- ► Method: find $f'(x)$ → set $f'(x)=0$ → test sign in each interval

6.3 Maxima & Minima

First Derivative Test:

- ► $f'(x)$ changes + to − at x=c $\Longrightarrow$ LOCAL MAXIMUM
- ► $f'(x)$ changes − to + at x=c $\Longrightarrow$ LOCAL MINIMUM
- ► $f'(x)$ does not change sign $\Longrightarrow$ NEITHER (inflection point)

Second Derivative Test:

- ► $f'(c) = 0$ and $f''(c) < 0$ $\Longrightarrow$ LOCAL MAXIMUM at x=c
- ► $f'(c) = 0$ and $f''(c) > 0$ $\Longrightarrow$ LOCAL MINIMUM at x=c
- ► $f'(c) = 0$ and $f''(c) = 0$ $\Longrightarrow$ TEST FAILS → use 1st derivative test

Absolute Extrema on Closed Interval [a,b]:

- ► Step 1: Find ALL critical points in (a,b)
- ► Step 2: Evaluate f at critical points AND endpoints f(a), f(b)
- ► Step 3: Largest value = Absolute Max; Smallest = Absolute Min

📌 Optimization Strategy

- Step 1: Identify objective function (quantity to max/min).
- Step 2: Express in ONE variable using the given constraint.
- Step 3: Find valid domain (e.g. length > 0, angle $\in (0,\pi)$).
- Step 4: Differentiate, set = 0, find critical point.
- Step 5: Confirm max/min via 2nd derivative test. State answer with units.

⚠ Common Mistakes

- Absolute max/min: ALWAYS check endpoints on closed interval.
- $f''(c) = 0$ means 2nd test fails — use 1st derivative test instead.
- Rate of change: distinguish dy/dx (space) from dy/dt (time).

6.4 Worked Example

Worked Example

Q: Find absolute max and min of $f(x)=2x^3-15x^2+36x+1$ on [1,5].

Step 1: $f'(x) = 6x^2-30x+36 = 6(x-2)(x-3)$; critical points in (1,5): $x=2, 3$

Step 2: $f(1)=24$, $f(2)=29$, $f(3)=28$, $f(5)=56$

∴ **Answer: Absolute Max = 56 at x=5; Absolute Min = 24 at x=1**

6.5 PYQ Triggers

Year	Board Exam Question
2024	Find point on $y=x^2-2x+3$ where tangent is parallel to x-axis. [2 marks]
2023	Find local max and min of $f(x)=\sin x-\cos x$ on $[0,2\pi]$. [5 marks]
2022	Ladder 5m leans against wall; top slides at 1/3 m/s. Find rate of angle change. [3 marks]
2021	Show height of cylinder of max volume in sphere of radius r is $2r/\sqrt{3}$. [5 marks]
2019	Find intervals where $f(x)=3x^4-4x^3-12x^2+5$ is strictly increasing/decreasing. [3 marks]

CHAPTER 7

Integrals

Standard formulas | Substitution | IBP | Partial Fractions | Definite integrals | Properties

7.1 Standard Integral Formulas

∫f(x)dx	Result (+C)	∫f(x)dx	Result (+C)
$\int x^n\,dx$	$x^{n+1}/(n+1)$, $n \neq -1$	$\int 1/x\,dx$	$\ln\lvert x\rvert$
$\int e^x\,dx$	e^x	$\int a^x\,dx$	$a^x / \ln a$
$\int \sin x\,dx$	$-\cos x$	$\int \cos x\,dx$	$\sin x$
$\int \tan x\,dx$	$\ln\lvert\sec x\rvert$	$\int \cot x\,dx$	$\ln\lvert\sin x\rvert$
$\int \sec x\,dx$	$\ln\lvert\sec x + \tan x\rvert$	$\int \operatorname{cosec} x\,dx$	$\ln\lvert\operatorname{cosec} x - \cot x\rvert$
$\int \sec^2 x\,dx$	$\tan x$	$\int \operatorname{cosec}^2 x\,dx$	$-\cot x$
$\int \sec x \tan x\,dx$	$\sec x$	$\int \operatorname{cosec} x \cot x\,dx$	$-\operatorname{cosec} x$
$\int 1/\sqrt{1-x^2}\,dx$	$\sin^{-1}x$	$\int 1/(1+x^2)\,dx$	$\tan^{-1}x$
$\int 1/\sqrt{a^2-x^2}\,dx$	$\sin^{-1}(x/a)$	$\int 1/(a^2+x^2)\,dx$	$(1/a)\tan^{-1}(x/a)$
$\int 1/(x^2-a^2)\,dx$	$(1/2a)\ln\lvert(x-a)/(x+a)\rvert$	$\int 1/(a^2-x^2)\,dx$	$(1/2a)\ln\lvert(a+x)/(a-x)\rvert$
$\int 1/\sqrt{x^2+a^2}\,dx$	$\ln\lvert x+\sqrt{x^2+a^2}\rvert$	$\int 1/\sqrt{x^2-a^2}\,dx$	$\ln\lvert x+\sqrt{x^2-a^2}\rvert$
$\int \sqrt{a^2-x^2}\,dx$	$x/2\cdot\sqrt{a^2-x^2}+a^2/2\cdot\sin^{-1}(x/a)$	$\int \sqrt{x^2+a^2}\,dx$	$x/2\cdot\sqrt{x^2+a^2}+a^2/2\cdot\ln\lvert x+\sqrt{x^2+a^2}\rvert$

7.2 Integration by Parts (IBP)

- ▶ $\int u \cdot v\,dx = u \cdot \int v\,dx - \int \left(\frac{du}{dx} \cdot \int v\,dx\right) dx$
- ▶ ILATE Rule (choose u as): Inverse trig > Log > Algebraic > Trig > Exponential
- ▶ ★ $\int e^x[f(x) + f'(x)]dx = e^x f(x) + C$ ← *Most important special case!*
- ▶ $\int \ln x\,dx = x \ln x - x + C$

7.3 Partial Fractions

Integrand Form	Decomposition
$(px+q) / [(x-a)(x-b)]$, $a \neq b$	$A/(x-a) + B/(x-b)$
$(px+q) / (x-a)^2$	$A/(x-a) + B/(x-a)^2$
$(px^2+qx+r) / [(x-a)(x^2+bx+c)]$	$A/(x-a) + (Bx+C)/(x^2+bx+c)$ [irreducible quadratic]

7.4 Definite Integrals — Key Properties

- ▶ **Fundamental Theorem:** $\int_a^b f(x)dx = F(b) - F(a)$ where $F'(x) = f(x)$
- ▶ ★ **KING'S PROPERTY:** $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ ← Appears EVERY year!
- ▶ Even function: $\int_{-a}^a f(x)dx = 2\int_0^a f(x)dx$ if $f(-x) = f(x)$
- ▶ Odd function: $\int_{-a}^a f(x)dx = 0$ if $f(-x) = -f(x)$

💡 King's Property — Classic Board Technique

- To find $\int_0^{\pi/2} \sin x/(\sin x+\cos x)\,dx$, call it I.
- By King's: $I = \int_0^{\pi/2} \cos x/(\cos x+\sin x)\,dx$ [substitute $x \to \pi/2-x$]
- Add: $2I = \int_0^{\pi/2} 1\,dx = \pi/2 \implies I = \pi/4$
- This technique appears in almost every Board paper!

⚠ Common Mistakes

- ALWAYS write +C for indefinite integrals. Missing C = mark loss.
- Definite integral substitution: MUST change limits to u-values.
- In IBP: wrong choice of u (ignoring ILATE) makes the integral harder.
- $\int_0^{2\pi} \sin x\,dx = 0 \neq$ area. Use $|\sin x|$ for area problems.

7.5 Worked Example

Worked Example

Q: Evaluate $\int_0^{\pi/2} \sin^2 x\,dx$

Step 1: Use identity: $\sin^2 x = (1 - \cos 2x)/2$
Step 2: $= (1/2)\int_0^{\pi/2}(1-\cos 2x)dx = (1/2)[x - \sin 2x/2]_0^{\pi/2}$
Step 3: $= (1/2)[(\pi/2 - 0) - (0 - 0)] = \pi/4$

∴ **Answer: $\pi/4$**

7.6 PYQ Triggers

Year	Board Exam Question
2024	Evaluate: $\int_0^{\pi/2} x/(\sin x+\cos x)\,dx$ [5 marks]
2023	Find: $\int (x+1)(x+\log x)^2/x\,dx$ [3 marks]
2022	Evaluate: $\int_{-1}^{1} |x^3-x|\,dx$ [3 marks]
2021	Integrate: $\int 1/(\sin^4 x+\cos^4 x)\,dx$ [5 marks]
2019	Evaluate: $\int_0^{\pi} (x \sin x)/(1+\cos^2 x)\,dx$ [5 marks]

CHAPTER 8

Application of Integrals

Area under curve | Area between curves | Standard areas

8.1 Area Formulas

- ► Area between y=f(x) and x-axis (a to b): $A = \int_a^b |f(x)|\, dx$
- ► Area between two curves [f(x) ≥ g(x) on (a,b)]: $A = \int_a^b [f(x) - g(x)]\, dx$
- ► Area of circle $x^2+y^2=r^2 = \pi r^2$ (verify: $4\int_0^r \sqrt{r^2-x^2}\, dx = \pi r^2$)
- ► Area of ellipse $x^2/a^2 + y^2/b^2 = 1 = \pi ab$

Strategy for Area Problems

- Step 1: Draw a rough sketch — essential for identifying upper/lower curves.
- Step 2: Find intersection points of two curves — these become integration limits.
- Step 3: Identify upper curve f(x) and lower curve g(x) on the interval.
- Step 4: Set up $A = \int[\text{upper} - \text{lower}]\, dx$ with correct limits.
- Step 5: If curve goes below x-axis, split at x-intercepts and use |f(x)|.
- Step 6: Area is ALWAYS positive.

8.2 Worked Example

Worked Example

Q: Find area enclosed between $y=x^2$ and $y=x$.

Step 1: Intersections: $x^2=x \rightarrow x=0, x=1$

Step 2: On [0,1]: line $y=x$ is above parabola $y=x^2$

Step 3: $A = \int_0^1(x-x^2)dx = [x^2/2 - x^3/3]_0^1 = 1/2 - 1/3 = 1/6$

∴ **Answer: Area = 1/6 sq. units**

8.3 PYQ Triggers

Year	Board Exam Question
2024	Find area of region bounded by $y=x$ and $y=x^3$. [3 marks]
2023	Find area: $\{(x,y): 0\le y\le x^2+1, 0\le y\le x+1, 0\le x\le 2\}$. [5 marks]
2022	Area bounded by $y=\lvert x+1\rvert$ and x-axis between $x=-4$ and $x=2$. [3 marks]
2021	Using integration, find area of △ with vertices (−1,0),(1,3),(3,2). [5 marks]
2019	Area of smaller region bounded by $x^2/9+y^2/4=1$ and $x/3+y/2=1$. [5 marks]

CHAPTER 9

Differential Equations

Order & Degree | Variable Separable | Homogeneous | Linear DE

🗑 **REMOVED FROM 2025-26 SYLLABUS (Official CBSE)**

- Bernoulli's Equation — NOT in 2025-26 CBSE syllabus

9.1 Order & Degree

- ► Order = highest derivative present in the equation
- ► Degree = power of the highest-order derivative (only when equation is polynomial form)
- ► Degree is UNDEFINED if highest derivative appears inside sin/cos/log/exp

Equation	Order	Degree
$y' = \sin x$	1	1
$(y')^2+y=x$	1	2
$y''+3y'+2y=0$	2	1
$y''+\sin(y')=0$	2	Undefined
$[1+(y')^2]^{(3/2)}=y''$	2	2

9.2 Variable Separable

- ► Form: $f(x)dx = g(y)dy \rightarrow \int g(y)dy = \int f(x)dx + C$

9.3 Homogeneous DE

- ► Homogeneous: $dy/dx = F(y/x)$ [numerator & denominator have equal degree]
- ► Substitute $y = vx \rightarrow dy/dx = v + x\cdot dv/dx$
- ► Converts to variable separable. Solve, then back-substitute $v = y/x$

9.4 Linear Differential Equation ★

- ► **Standard form:** $\frac{dy}{dx} + P(x)\cdot y = Q(x)$
- ► **Integrating Factor (IF) =** $e^{\int P(x)dx}$
- ► **Solution:** $y\cdot(IF) = \int Q\cdot(IF)\,dx + C$

⚠ **Common Mistakes**

- Order ≠ degree. Degree is undefined when highest derivative is in trig/log/exp.
- Standard form: coefficient of dy/dx must be 1. Divide through first if needed.
- Always add +C. In homogeneous: back-substitute v=y/x at the end.

9.5 Worked Example

✏ Worked Example

Q: Solve linear DE: $dy/dx + (2/x)y = x^2$

Step 1: $P(x) = 2/x$, $Q(x) = x^2$
Step 2: IF = $e^{\int 2/x\, dx} = e^{2 \ln x} = x^2$
Step 3: Multiply: $d/dx[y \cdot x^2] = x^2 \cdot x^2 = x^4$
Step 4: Integrate: $y \cdot x^2 = x^5/5 + C$

∴ **Answer: $y = x^3/5 + C/x^2$**

9.6 PYQ Triggers

Year	Board Exam Question
2024	Solve: $(x^2 - yx^2)dy + (y^2 + xy^2)dx = 0$ [3 marks]
2023	Solve: $dy/dx + 2y \tan x = \sin x$, given $y=0$ at $x=\pi/3$. [5 marks]
2022	Solve: $(x+y)\, dy/dx = 1$ [3 marks]
2021	Solve: $(1+x^2)dy + 2xy\, dx = \cot x\, dx$ [3 marks]
2019	Solve: $x\, dy/dx + y = x \cos x + \sin x$, given $y(\pi/2)=1$. [5 marks]

CHAPTER 10

Vector Algebra

Types | Dot product | Cross product | Area formulas

🗑 **REMOVED FROM 2025-26 SYLLABUS (Official CBSE)**

- Scalar Triple Product — NOT in 2025-26 syllabus (only dot & cross product tested)

10.1 Types of Vectors

Type	Definition
Zero Vector ($\vec{0}$)	Magnitude = 0
Unit Vector (â)	$\lvert\hat{a}\rvert = 1$, $\hat{a} = \vec{a}/\lvert\vec{a}\rvert$, Standard: $\hat{i},\hat{j},\hat{k}$
Collinear/Parallel	$\vec{a} \times \vec{b} = \vec{0}$
Position Vector	$\vec{OP}$ = position of P from origin O

10.2 Magnitude & Section Formula

- If $\vec{a} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$: $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$
- Section formula (internal, m:n): $\vec{OP} = (m\cdot\vec{OB} + n\cdot\vec{OA})/(m+n)$
- Midpoint: $\vec{OM} = (\vec{OA} + \vec{OB})/2$

10.3 Dot (Scalar) Product

- $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta \quad \to \quad \cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$
- $\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$ (component form)
- $\hat{\imath}\cdot\hat{\imath} = \hat{\jmath}\cdot\hat{\jmath} = \hat{k}\cdot\hat{k} = 1$ | $\hat{\imath}\cdot\hat{\jmath} = \hat{\jmath}\cdot\hat{k} = \hat{k}\cdot\hat{\imath} = 0$
- $\vec{a} \perp \vec{b}$ iff $\vec{a}\cdot\vec{b} = 0$ (both non-zero vectors)
- Scalar projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$ | Vector projection $= \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2} \cdot \vec{b}$

10.4 Cross (Vector) Product

- $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta \cdot \hat{n}$ ($\hat{n}$ perpendicular to both, right − hand rule)
- $\vec{a} \times \vec{b} = |\hat{\imath}\ \hat{\jmath}\ \hat{k}| = (a_2b_3 - a_3b_2)\hat{\imath} - (a_1b_3 - a_3b_1)\hat{\jmath} + (a_1b_2 - a_2b_1)\hat{k}$
- Anti-commutative: $\vec{a}\times\vec{b} = -(\vec{b}\times\vec{a})$
- $\hat{\imath}\times\hat{\imath} = \hat{\jmath}\times\hat{\jmath} = \hat{k}\times\hat{k} = \vec{0}$ | $\hat{\imath}\times\hat{\jmath} = \hat{k}$, $\hat{\jmath}\times\hat{k} = \hat{\imath}$, $\hat{k}\times\hat{\imath} = \hat{\jmath}$ (cyclic)
- $\vec{a} \parallel \vec{b}$ iff $\vec{a}\times\vec{b} = \vec{0}$
- Area of parallelogram $= |\vec{a} \times \vec{b}|$ Area of triangle $= \frac{1}{2}|\vec{a} \times \vec{b}|$

⚠ **Common Mistakes**

- $\vec{a}\cdot\vec{b}$ = SCALAR. $\vec{a}\times\vec{b}$ = VECTOR. Never confuse these.
- $\vec{a}\times\vec{b} = -(\vec{b}\times\vec{a})$. NOT commutative.
- $\hat{\imath}\times\hat{\imath} = \vec{0}$, NOT 1. Only $\hat{\imath}\cdot\hat{\imath} = 1$.

- Area of triangle = HALF of |cross product|.

10.5 Worked Example

Worked Example

Q: Find unit vector perpendicular to both $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$ and $\vec{b}=-\hat{i}+2\hat{j}-\hat{k}$.

Step 1: $\vec{a}\times\vec{b}$ = det[$\hat{i}$ $\hat{j}$ $\hat{k}$/1 2 3/−1 2 −1] = $(-2-6)\hat{i}-(-1+3)\hat{j}+(2+2)\hat{k} = -8\hat{i}-2\hat{j}+4\hat{k}$

Step 2: $|\vec{a}\times\vec{b}| = \sqrt{64+4+16} = \sqrt{84} = 2\sqrt{21}$

Step 3: Unit vector = $(-8\hat{i}-2\hat{j}+4\hat{k})/(2\sqrt{21}) = (-4\hat{i}-\hat{j}+2\hat{k})/\sqrt{21}$

∴ **Answer: $\hat{n} = (-4\hat{i} - \hat{j} + 2\hat{k})/\sqrt{21}$**

10.6 PYQ Triggers

Year	Board Exam Question
2024	If $\vec{a},\vec{b}$ unit vectors with angle θ, find $|\vec{a}-\vec{b}|$. [2 marks]
2023	Find projection of $\vec{a}=2\hat{i}+3\hat{j}+2\hat{k}$ on $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$. [2 marks]
2022	If $\vec{a}+\vec{b}+\vec{c}=0$, $|\vec{a}|=3$, $|\vec{b}|=5$, $|\vec{c}|=7$, find angle between $\vec{a}$ and $\vec{b}$. [3 marks]
2021	Find λ so that $(2\hat{i}+6\hat{j}+14\hat{k})$ and $(\hat{i}-\lambda\hat{j}+7\hat{k})$ are perpendicular. [1 mark]
2019	Find unit vector perpendicular to both $3\hat{i}+2\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}+3\hat{k}$. [3 marks]

CHAPTER 11

Three Dimensional Geometry

Direction cosines | Lines in 3D | Skew lines | Shortest Distance | Angle between lines

REMOVED FROM 2025-26 SYLLABUS (Official CBSE)

- ENTIRE section on Planes — REMOVED from 2025-26 official syllabus
- Angle between two planes, Distance from point to plane — REMOVED
- Angle between line and plane, Foot of perpendicular to plane — REMOVED
- Only LINES in 3D are in the 2025-26 syllabus

11.1 Direction Cosines & Ratios

- $l^2 + m^2 + n^2 = 1 \quad \leftarrow$ always true for direction cosines (NOT for direction ratios)
- If DRs are (a, b, c): DCs = $\left(\frac{a}{d}, \frac{b}{d}, \frac{c}{d}\right)$ where $d = \sqrt{a^2 + b^2 + c^2}$
- DRs of line joining (x_1,y_1,z_1) and (x_2,y_2,z_2): $(x_2-x_1, y_2-y_1, z_2-z_1)$
- Parallel lines: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ Perpendicular: $a_1a_2 + b_1b_2 + c_1c_2 = 0$
- Perpendicular iff $a_1a_2 + b_1b_2 + c_1c_2 = 0$
- Angle between two lines: $\cos\theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2+b_1^2+c_1^2}\cdot\sqrt{a_2^2+b_2^2+c_2^2}}$

11.2 Equations of a Line in 3D

- Vector form: $\vec{r} = \vec{a} + \lambda\vec{b}$ ($\vec{a}$ = point on line, $\vec{b}$ = direction vector, $\lambda \in \mathbb{R}$)
- Cartesian form: $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$
- Through two points: $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$

11.3 Shortest Distance between Lines

- Skew lines SD: $\text{SD} = \frac{|(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$
- Parallel lines SD: $\text{SD} = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$
- Lines INTERSECT iff SD = 0

⚠ Common Mistakes

- $l^2+m^2+n^2=1$ for DCs only — NOT for direction ratios.
- Angle between lines: ALWAYS take |dot product| in the numerator.
- If $\vec{b}_1\times\vec{b}_2=\vec{0}$ (parallel): use parallel SD formula, not the skew formula.

11.4 Worked Example

Worked Example

Q: Find SD between $\vec{r}=(\hat{i}+2\hat{j}+\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=(2\hat{i}-\hat{j}-\hat{k})+\mu(2\hat{i}+\hat{j}+2\hat{k})$.

Step 1: $\vec{a}_2-\vec{a}_1 = \hat{i}-3\hat{j}-2\hat{k}$

Step 2: $\vec{b}_1\times\vec{b}_2 = \det[\hat{i}\ \hat{j}\ \hat{k}/1\ -1\ 1/2\ 1\ 2] = -3\hat{i}+0\hat{j}+3\hat{k}$, $|\vec{b}_1\times\vec{b}_2| = 3\sqrt{2}$

Step 3: $(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2) = (1)(-3)+(-3)(0)+(-2)(3) = -9$

Step 4: $SD = |-9|/(3\sqrt{2}) = 9/(3\sqrt{2}) = 3/\sqrt{2}$

∴ **Answer: SD = $3\sqrt{2}/2$**

11.5 PYQ Triggers

Year	Board Exam Question
2024	Find angle between lines $(x-2)/3=(y+1)/-2=(z-3)/6$ and $(x-2)/-1=(y+1)/2=z/2$. [2 marks]
2023	Find SD between $\vec{r}=(\hat{i}+2\hat{j}+\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=(2\hat{i}-\hat{j}-\hat{k})+\mu(2\hat{i}+\hat{j}+2\hat{k})$. [5 marks]
2022	Find image of point (1,6,3) in line $x/1=(y-1)/2=(z-2)/3$. [5 marks]
2021	Find foot of perpendicular from (1,2,3) to line $x/3=(y+1)/-1=(z-3)/5$. [3 marks]
2019	Find equation of line passing through (2,1,3) perpendicular to $x/1=y/2=z/3$ and $x/-3=y/2=z/5$. [3 marks]

CHAPTER 12

Linear Programming

Formulation | Graphical Method | Corner Point Theorem | ≤ 3 constraints

12.1 Key Terms

Term	Definition
Objective Function	Z = ax+by — quantity to be maximized or minimized
Decision Variables	x and y — unknowns to determine
Constraints	Linear inequalities restricting x and y
Non-negativity	$x \geq 0$, $y \geq 0$ — ALWAYS present
Feasible Region	All points satisfying ALL constraints simultaneously
Corner Point	Vertex of the feasible polygon
Bounded Region	Enclosed on all sides (finite area)
Unbounded Region	Extends infinitely in some direction

12.2 Corner Point Theorem & Method

- ► Optimal solution (if it exists) occurs at a CORNER POINT of the feasible region
- ► Bounded region: Both max and min exist at corner points
- ► Unbounded region: Must verify using open half-plane test

🧠 Complete 8-Step Graphical Method

- Step 1: Formulate Z=ax+by and all constraints (including $x,y \geq 0$).
- Step 2: Plot each constraint line (x-intercept: y=0; y-intercept: x=0).
- Step 3: Determine feasible side of each line — test point (0,0).
- Step 4: Feasible region = intersection of ALL feasible sides.
- Step 5: Find ALL corner points (solve pairs of boundary lines).
- Step 6: Evaluate Z at EVERY corner point.
- Step 7: Largest Z → maximum; Smallest Z → minimum.
- Step 8: State x, y, and optimal Z with units.

⚠ Common Mistakes

- $x \geq 0$, $y \geq 0$ ALWAYS apply — never forget these constraints.
- Test (0,0): if it satisfies constraint → shade toward origin.
- Must find ALL corner points, including intersections of non-trivial constraints.
- For unbounded region: MUST verify using the open half-plane test.

12.3 Worked Example

Worked Example

Q: Maximize $Z=3x+4y$ s.t. $x+y\le4$, $x+2y\le6$, $x,y\ge0$.

Step 1: Corner points: O(0,0), A(4,0), B(2,2), C(0,3)
Step 2: Find B: solve $x+y=4$ and $x+2y=6$ → $y=2$, $x=2$
Step 3: Z(O)=0, Z(A)=12, Z(B)=14, Z(C)=12

∴ **Answer: Maximum Z = 14 at (2, 2)**

12.4 PYQ Triggers

Year	Board Exam Question
2024	Maximize $Z=x+2y$ s.t. $x+2y\le100$, $2x+y\le150$, $x+y\ge20$, $x,y\ge0$. [5 marks]
2022	Minimize $Z=13x-15y$ s.t. $x+y\le7$, $2x-3y+6\ge0$, $x,y\ge0$. [5 marks]
2021	Maximize $Z=3x+2y$ s.t. $x+y\le4$, $x-y\le2$, $x,y\ge0$. [5 marks]
2019	Diet has ≥80 units Vitamin A and ≤120 units minerals. Minimize cost. [5 marks]

CHAPTER 13

Probability

Conditional probability | Independence | Total probability | Bayes' Theorem

REMOVED FROM 2025-26 SYLLABUS (Official CBSE)

- Random Variables and Probability Distribution — REMOVED from 2025-26
- Binomial Distribution — REMOVED from 2025-26
- Mean and Variance of distributions — REMOVED
- Only: Conditional probability, multiplication theorem, independence, total probability, Bayes' theorem

13.1 Fundamental Probability Formulas

- ▶ $P(A) \in [0,1]$ | $P(S) = 1$ | $P(\varphi) = 0$ | $P(A') = 1 - P(A)$
- ▶ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- ▶ Mutually exclusive: $P(A \cap B) = 0 \rightarrow P(A \cup B) = P(A) + P(B)$
- ▶ **Conditional Probability:** $P(A|B) = \frac{P(A \cap B)}{P(B)}$ $[P(B) > 0]$
- ▶ Multiplication Rule: $P(A \cap B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$
- ▶ Independent Events: $P(A \cap B) = P(A) \cdot P(B)$ ← DEFINITION of independence
- ▶ If independent: $P(A|B) = P(A)$ and $P(B|A) = P(B)$

13.2 Total Probability & Bayes' Theorem ★

- ▶ **Total Probability:** $P(A) = \Sigma P(E_i) \cdot P(A|E_i)$
- ▶ **Bayes' Theorem:** $P(E_k|A) = \frac{P(E_k) \cdot P(A|E_k)}{\Sigma\ P(E_i) \cdot P(A|E_i)}$

Bayes' Theorem — How to Solve

- Step 1: Draw a tree diagram for ALL given information.
- Step 2: List hypotheses E_1, E_2, ... with PRIOR probabilities $P(E_i)$.
- Step 3: List conditional probabilities $P(A|E_i)$ for each hypothesis.
- Step 4: $P(A) = \Sigma\ P(E_i) \cdot P(A|E_i)$ [Total Probability formula].
- Step 5: Apply Bayes' formula for required posterior $P(E_k|A)$.

⚠ Common Mistakes

- ME ≠ Independent. ME: $P(A \cap B)=0$. Indep: $P(A \cap B)=P(A)P(B)$.
- $P(A|B) \neq P(B|A)$ unless $P(A)=P(B)$. Students swap these often.
- ME events with positive probability CANNOT be independent.
- Complement: P(at least 1) = 1−P(none). Always faster than direct calculation.

13.3 Worked Example

Worked Example

Q: Machines A,B,C produce 50%,30%,20% of output; defect rates 1%,2%,3%. Item found defective. Find P(from machine A).

Step 1: $P(A)=0.5$, $P(B)=0.3$, $P(C)=0.2$; $P(D|A)=0.01$, $P(D|B)=0.02$, $P(D|C)=0.03$

Step 2: $P(D) = 0.5\times0.01 + 0.3\times0.02 + 0.2\times0.03 = 0.005+0.006+0.006 = 0.017$

Step 3: $P(A|D) = (0.5\times0.01)/0.017 = 0.005/0.017$

∴ **Answer: P(from machine A | defective) = 5/17**

13.4 PYQ Triggers

Year	Board Exam Question
2024	A and B throw die alternately. A wins on 6, B wins on 5 or 6. Find winning probabilities. [5 marks]
2023	Machines A:30%,B:50%,C:20%. Defect rates 1%,1.5%,0.5%. P(from B|defective)? [5 marks]
2022	$P(A)=0.6$, $P(B)=0.5$, $P(A\cap B)=0.3$. Find $P(A\|B)$ and $P(B\|A)$. Are A,B independent? [3 marks]
2021	Bag has 5R,4W balls. 2 drawn. Find P(both same colour). [2 marks]
2019	$P(A)=0.4$, $P(B)=0.3$, $P(A\cap B)=0.1$. Find $P(A\cup B)$. Are A,B independent? [3 marks]

Master Quick Revision Sheet — All 13 Chapters (2025-26)

Chapter	Key Formulas (2025-26 syllabus only)
Ch 1 Relations	Equiv = Reflexive+Sym+Trans | f bijective ↔ f^{-1} exists
Ch 2 ITF	$\sin^{-1}x+\cos^{-1}x=\pi/2$ | $\tan^{-1}(x+y)/(1-xy)$ | $2\tan^{-1}x=\sin^{-1}[2x/(1+x^2)]$
Ch 3 Matrices	$(AB)^T=B^TA^T$ (REVERSAL) | $A=½(A+A^T)+½(A-A^T)$
Ch 4 Determinants	$A^{-1}=adj(A)/\|A\|$ | $X=A^{-1}B$ | $\|adjA\|=\|A\|^{(n-1)}$
Ch 5 C&D	Diff⟹Cont | $d/dx[a^x]=a^x\ln a$ | $d^2y/dx^2=[d(dy/dx)/dt]/(dx/dt)$ for parametric
Ch 6 App.Deriv.	Inc: $f'>0$ | Max: $f'=0, f''<0$ | Abs.max/min: check endpoints
Ch 7 Integrals	ILATE | $\int e^x[f+f']=e^xf(x)+C$ | King's: $\int_0^a f(x)=\int_0^a f(a-x)dx$
Ch 8 App.Integrals	A=∫[upper−lower]dx | Sketch first | Area always positive
Ch 9 Diff.Eqs.	Sep.var | Homogeneous: y=vx | Linear: $IF=e^{\int Pdx}$, $y(IF)=\int Q(IF)dx+C$
Ch 10 Vectors	$\vec{a}\cdot\vec{b}=\|a\|\|b\|\cos\theta$ (scalar) | $\vec{a}\times\vec{b}=\|a\|\|b\|\sin\theta\hat{n}$ | Area $\triangle=½\|\vec{a}\times\vec{b}\|$
Ch 11 3D Lines	$l^2+m^2+n^2=1$ | $SD=\|(a_2-a_1)\cdot(b_1\times b_2)\|/\|b_1\times b_2\|$
Ch 12 LPP	Optimal at corner point | Up to 3 constraints | Check ALL vertices
Ch 13 Prob.	$P(A\|B)=P(A\cap B)/P(B)$ | Bayes: $P(E_k\|A)=P(E_k)P(A\|E_k)/\Sigma P(E_i)P(A\|E_i)$

Examiner Notes & Board Exam Strategy 2025-26

Strategy	What to Do
Step-by-step	Each step on a new line. Box final answer. Marks given per step.
State formula first	'By Bayes' theorem:...' earns 1 mark even if calculation has errors.
Never leave blank	Write formula even if unsure — partial marks always apply.
Diagram = marks	In Ch8 (areas), Ch11 (3D): sketch = 1 direct mark.
Vector notation	Arrows on vectors, matrices in [ ]. Notation errors cost marks.
Time per question	1-mark: 1.5 min | 2-mark: 3 min | 3-mark: 5 min | 5-mark: 8 min
High priority	Integrals (Ch7+8) ~18 marks | Calculus ~15 marks | Vectors+3D ~14 marks
King's Property	$\int_0^a f(x)=\int_0^a f(a-x)dx$ — appears EVERY year. MUST master.
Complement method	P(at least 1) = 1−P(none). ALWAYS faster than direct sum.
DO NOT study	Tangents/Normals | Planes | Binomial Dist. | LMVT | Bernoulli | Triple product

📌 Final Note from SJ Maths

- This handbook covers ONLY the official 2025-26 CBSE syllabus — nothing extra.
- Deleted: Tangents/Normals | Planes | Binomial | LMVT | Bernoulli | Scalar Triple Product

- Visit sjmaths.com for FREE video solutions to every PYQ listed in this book.
- Best wishes for your 2025-26 CBSE Board Exams! 🎯